DO NOW — RECALL

Unfamiliar algorithm design and evaluation

Name: _____ Class: _____ Date: _____

1. What is a precondition?

2. Why write edge-case tests?

3. What does a loop invariant express?

4. C# array [4,1,4,2]: compute sum at indexes 0 and 2, and number of distinct values.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Requirement for an algorithm to work correctly.

2. Model answer

To expose faults on boundaries/unusual inputs.

3. Model answer

A property that stays true after each loop iteration.

4. Model answer

Indexes 0 and 2 sum8; distinct values {4,1,2} count3.

By the end, you can…

• State the precondition

• Trace the invariant

• Evaluate and test

Vocabulary — know these terms

invariant — Condition maintained throughout a loop.

precondition — Assumption required for correctness.

boundary test — Tests around value/size limits.

time complexity — Growth in work as input increases.

NEW LEARNING

Read, annotate and explain

State the precondition

The input must be sorted; equal values then appear consecutively.

Trace the invariant

Start with one distinct value for a nonempty array. Each change from previous value adds one.

Evaluate and test

The algorithm scans once O(n) time, O(1) extra space; test empty, all equal and all distinct inputs.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Trace distinct count in [2,2,5,5,7,7,7].

Predict first

Duplicates do not increment the counter because each equals the preceding element.

Explain the method

Only changes to five and seven increment count; result is three distinct values.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Trace DistinctCount on [3,3,3,6,8,8]. Record count after each index. [6]

Question 2

Describe a correct set of test cases for the algorithm and their expected outputs. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why the algorithm may fail on unsorted [2,1,2]. Give a repair and complexity consequence. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

The distinct-count algorithm is valid for unsorted inputs without modification.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.